Game Theory

Final Examination: Solutions and Grading Rubric | Fall 2023

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Grading notes for GSIs

- Total 100 points. Use the rubric boxes; deduct an arithmetic slip only once and carry the error forward.
- Decimal answers that are exactly equal to the fraction (e.g. 0.4 for $\frac{2}{5}$) receive full credit. Rounded decimals for surds (e.g. 0.457) receive full credit if the exact expression also appears in the work.

Problem 1 A 3×3 Zero-Sum Game [18 points]

	c_1	c_2	c_3
r_1	4	-1	5
r_2	-2	3	-1
r_3	0	0	-1

(a) r_1 does not dominate r_3 ($-1 < 0$ against c_2); r_2 does not dominate r_3 ($-2 < 0$ against c_1). For the mixture,

$$\alpha r_1 + (1-\alpha) r_2 = \big(6\alpha - 2,\ 3 - 4\alpha,\ 6\alpha - 1\big) > (0,\, 0,\, -1) \iff \alpha > \tfrac{1}{3},\ \alpha < \tfrac{3}{4},\ \alpha > 0.$$

So r_3 is strictly dominated exactly for $\boxed{\alpha \in (\tfrac{1}{3}, \tfrac{3}{4})}$; for instance $\alpha = \frac{1}{2}$ gives $(1, 1, 2)$.

Rubric

1 pt: rules out both pure strategies. 1 pt: mixture payoff vector. 2 pts: interval $(\frac{1}{3}, \frac{3}{4})$. One valid α with verification but no interval: 1 of these 2 pts.

(b) Without r_3, column $c_1 = (4, -2)$ and column $c_3 = (5, -1)$: c_3 costs Column more in both rows, so c_3 is strictly dominated by c_1. In the original game c_3 is *better* for Column than c_1 against r_3 ($-1 < 0$), and against r_3 every mixture of c_1, c_2 pays $0 > -1$; so c_3 was not dominated (by a pure or mixed strategy) until r_3 was removed.

Rubric

2 pts: c_3 dominated by c_1 after deleting r_3, with the direction (Column minimizes) stated correctly. 1 pt: explanation using the r_3 row.

(c) Remaining game $\begin{pmatrix} 4 & -1 \\ -2 & 3 \end{pmatrix}$. Maximin in pure strategies is -1, minimax is 3, so there is no saddle point and both players mix.

- Row plays r_1 w.p. p: against c_1, $6p - 2$; against c_2, $3 - 4p$. Equalizing gives $p = \frac{1}{2}$ and $v = 1$.
- Column plays c_1 w.p. q: against r_1, $5q - 1$; against r_2, $3 - 5q$. Equalizing gives $q = \frac{2}{5}$ and $v = 1$.

Value $\boxed{v = 1}$; Row $\boxed{(\frac{1}{2}, \frac{1}{2})}$; Column $\boxed{(\frac{2}{5}, \frac{3}{5})}$.

Rubric

1 pt: checks for a saddle point. 2 pts: $p = \frac{1}{2}$. 2 pts: $q = \frac{2}{5}$. 2 pts: $v = 1$ (must be consistent from both sides).
Common error: attaching $\frac{2}{5}$ to the wrong column (reporting Column's strategy as $(\frac{3}{5}, \frac{2}{5})$). Deduct 2.

(d) In the original game: Row $x^* = (\frac{1}{2}, \frac{1}{2}, 0)$, Column $y^* = (\frac{2}{5}, \frac{3}{5}, 0)$, value 1.

$$x^{*\top} A = (1,\ 1,\ 2) \ \geq 1 \text{ in every column}, \qquad Ay^* = (1,\ 1,\ 0)^\top \ \leq 1 \text{ in every row}.$$

So Row guarantees at least 1 and Column holds Row to at most 1; both strategies are optimal and $v = 1$.

Rubric

1 pt: strategies padded with zeros. 3 pts: both verification vectors, including the payoff against the deleted strategies (2 against c_3 and 0 for r_3). Omitting the deleted strategies from the check: -1.

Problem 2 An Extensive Game with a Simultaneous-Move Subgame [22 points]

(a) Player 1 chooses an action at the root and an action in the subgame (even if the subgame is not reached): $\{(\mathsf{Out}, a), (\mathsf{Out}, b), (\mathsf{In}, a), (\mathsf{In}, b)\}$. Player 2 has one information set: $\{\ell, r\}$.

	ℓ	r
(Out, a)	2, 3	2, 3
(Out, b)	2, 3	2, 3
(In, a)	4, 1	0, 0
(In, b)	0, 0	1, 4

Rubric

2 pts: four strategies for Player 1 (1 pt if only $\{\mathsf{Out}, (\mathsf{In}, a), (\mathsf{In}, b)\}$ is given, a reduced normal form). 1 pt: Player 2's strategies. 2 pts: correct payoff table.

(b) *Subgame.* Pure equilibria (a, ℓ) with payoffs $(4, 1)$ and (b, r) with payoffs $(1, 4)$. Mixed: let $p = \Pr(a)$, $q = \Pr(\ell)$. Player 2 indifferent: $p \cdot 1 = (1 - p) \cdot 4 \Rightarrow p = \frac{4}{5}$. Player 1 indifferent: $4q = 1 - q \Rightarrow q = \frac{1}{5}$. Payoffs $(4q,\ 4(1 - p)) = (\frac{4}{5}, \frac{4}{5})$.

Root. Player 1 compares Out (payoff 2) with the continuation payoff of the subgame equilibrium.

Subgame NE	P1 at root	SPE	Outcome / payoffs
(a, ℓ): $(4, 1)$	$4 > 2$: In	$\big((\mathsf{In}, a),\ \ell\big)$	In, (a, ℓ); $(4, 1)$
(b, r): $(1, 4)$	$1 < 2$: Out	$\big((\mathsf{Out}, b),\ r\big)$	Out; $(2, 3)$
mixed: $(\frac{4}{5}, \frac{4}{5})$	$\frac{4}{5} < 2$: Out	$\big((\mathsf{Out}, \frac{4}{5}a + \frac{1}{5}b),\ \frac{1}{5}\ell + \frac{4}{5}r\big)$	Out; $(2, 3)$

These three are all the SPE.

Rubric

2 pts: two pure subgame NE. 3 pts: mixed subgame NE ($p = \frac{4}{5}$, $q = \frac{1}{5}$) with payoff $\frac{4}{5}$ to Player 1. 3 pts: correct root choice in each case (1 each). 1 pt: SPE written as complete strategy profiles (Player 1's subgame action specified even after Out).
Common error: writing the second SPE as "(Out, r)" without Player 1's subgame action: -1 once.

(c) $\big((\mathsf{Out}, a),\ r\big)$ is a Nash equilibrium: against r, Player 1 gets 2 from Out, 0 from (In, a) and 1 from (In, b), so Out is optimal; since Out ends the game, every strategy of Player 2 is a best response. It is not subgame perfect because its restriction to the subgame, (a, r), is not a Nash equilibrium of the subgame: Player 2 would switch to ℓ ($1 > 0$), and Player 1 would switch to b ($1 > 0$). Player 2's choice of r against a is a non-credible threat that only works because the subgame is never reached.

Rubric

2 pts: correct profile. 1 pt: verification that it is a NE. 2 pts: explanation that the subgame restriction (a, r) is not a NE.
Answers giving $\big((\mathsf{Out}, b), r\big)$ (which *is* an SPE) receive 0 for the profile and at most 1 pt total.

(d) If Player 2 commits to r, Player 1's best action after In is b (payoff 1), which is worse than Out (payoff 2); so Player 1 stays Out and Player 2 receives 3. In the SPE $\big((\mathsf{In}, a), \ell\big)$ Player 2 receives only 1; in the other two SPE she receives 3. Commitment therefore *weakly* benefits Player 2: it removes the equilibrium that is bad for her and guarantees 3.

Rubric

1 pt: Player 1 stays Out after commitment. 2 pts: comparison with all three SPE payoffs for Player 2.

Problem 3 Repeated Prisoner's Dilemma [20 points]

(a) D strictly dominates C for each player ($8 > 6$, $3 > 1$), so (D, D) is the unique stage NE. With T repetitions, in period T every SPE must play the unique stage NE after every history, since nothing follows. Given that, the continuation in period $T - 1$ does not depend on current actions, so period-$(T-1)$ play must also be the stage NE; by backward induction the unique SPE plays (D, D) after every history in every period.

Rubric

1 pt: (D, D) with dominance argument. 2 pts: backward-induction argument (1 pt if it only states "unraveling" without the reason that continuation play is history-independent).

(b) *Cooperative phase.* Conforming gives $\dfrac{6}{1-\delta}$. The best one-shot deviation is D, which yields 8 today and 3 forever after:

$$\frac{6}{1-\delta} \;\geq\; 8 + \frac{3\delta}{1-\delta} \iff 6 \geq 8(1-\delta) + 3\delta \iff \delta \geq \tfrac{2}{5}.$$

Punishment phase. Play is (D, D) forever regardless of what happens; deviating to C gives $1 < 3$ today and leaves the future unchanged, so there is no profitable deviation. By the one-shot deviation principle, grim trigger is an SPE iff $\boxed{\delta \geq \frac{2}{5}}$.

Rubric

1 pt: payoff from cooperating. 2 pts: payoff from deviating. 2 pts: $\delta \geq \frac{2}{5}$. 1 pt: punishment phase checked. Students who find $\frac{2}{5}$ without checking the punishment phase: 5/6.

(c) *Cooperative phase.* A deviation gains $8 - 6 = 2$ today, then the deviator receives 3 instead of 6 in each of the next k periods, after which cooperation resumes; later payoffs are identical. The deviation is unprofitable iff

$$2 \;\leq\; 3\,(\delta + \delta^2 + \cdots + \delta^k).$$

Punishment phase. Conforming to D gives 3 today; deviating to C gives 1 today and restarts the k-period punishment, so it delays the return to 6. Deviating is never profitable.

- $k = 1$: $2 \leq 3\delta \iff \boxed{\delta \geq \frac{2}{3}}$.
- $k = 2$: $3\delta^2 + 3\delta - 2 \geq 0 \iff \boxed{\delta \geq \dfrac{\sqrt{33} - 3}{6} \approx 0.457}$.

Rubric

2 pts: correct incentive inequality for general k (or for each case separately). 1 pt: punishment phase checked. 2 pts: $k = 1$ threshold. 2 pts: $k = 2$ threshold (1 pt if the quadratic is correct but the wrong root is chosen).

(d) $\frac{2}{5} < \frac{\sqrt{33}-3}{6} < \frac{2}{3}$: grim trigger (infinitely long punishment) needs the least patience, then $k = 2$, then $k = 1$. Longer punishments impose a larger loss on a deviator, so cooperation can be sustained with a smaller δ; as $k \to \infty$ the k-period threshold decreases to the grim-trigger threshold $\frac{2}{5}$.

Rubric

2 pts: correct ranking. 2 pts: explanation.

Problem 4 A Bayesian Game [22 points]

(a) Player 1: $\{U, D\}$. Player 2's pure strategy specifies an action for each type: $\{LL, LR, RL, RR\}$, where the first letter is type t_1's action. Type t_2: R gives $1 > 0$ against U and $3 > 0$ against D, so R is strictly dominant.

Rubric

1 pt: Player 1's strategies. 1 pt: Player 2's four type-contingent strategies. 1 pt: dominance for t_2.

(b) Type t_2 plays R. Type t_1's best response is L against U $(2 > 0)$ and R against D $(1 > 0)$.

- Candidate $(U;\, L, R)$: types are best-responding. Player 1: $u_1(U) = 5\mu$, $u_1(D) = \mu{\cdot}1 + (1-\mu){\cdot}2 = 2 - \mu$. Equilibrium iff $5\mu \geq 2 - \mu$, i.e. $\mu \geq \frac{1}{3}$.
- Candidate $(D;\, R, R)$: types are best-responding. Player 1: $u_1(U) = 0 < 2 = u_1(D)$. Equilibrium for every μ.
- Any other combination has type t_1 or t_2 not best-responding.

Pure BNE: $\boxed{(D; R, R) \text{ for all } \mu}$, and $\boxed{(U; L, R) \text{ iff } \mu \geq \frac{1}{3}}$.

Rubric

1 pt: type t_1's best responses. 3 pts: $(U; L, R)$ with the threshold $\mu \geq \frac{1}{3}$. 2 pts: $(D; R, R)$ for all μ.
1 pt: argument that there are no others.
Common error: treating Player 2 as a single player choosing one action (ignoring types): at most 3/7.

(c) Let $p = \Pr(U)$ and $a = \Pr(t_1 \text{ plays } L)$; type t_2 plays R.

- Type t_1 indifferent: $u(L) = 2p$, $u(R) = 1 - p$, so $p = \frac{1}{3}$.
- Player 1 indifferent: $\Pr(L) = \mu a = \frac{a}{2}$. Then $u_1(U) = 5 \cdot \frac{a}{2}$ and $u_1(D) = 1 \cdot \frac{a}{2} + 2\left(1 - \frac{a}{2}\right) = 2 - \frac{a}{2}$. Setting them equal: $3a = 2$, so $a = \frac{2}{3}$.

BNE: $\boxed{\Pr(U) = \frac{1}{3};\ \ t_1 \text{ plays } L \text{ w.p. } \frac{2}{3};\ \ t_2 \text{ plays } R}$.

Rubric

3 pts: $p = \frac{1}{3}$ from type t_1's indifference. 4 pts: $a = \frac{2}{3}$ from Player 1's indifference using the prior correctly. 1 pt: complete statement including t_2's action.
Using $\Pr(L) = a$ instead of μa (ignoring the prior) gives $a = \frac{1}{3}$: award 5/8.

(d) At $\mu = \frac{1}{2}$: $(U; L, R)$ gives $5\mu = \frac{5}{2}$; $(D; R, R)$ gives 2; the mixed BNE gives $u_1(U) = \frac{5}{2} \cdot \frac{2}{3} = \frac{5}{3}$. Player 1 prefers $\boxed{(U; L, R)}$.

Rubric

3 pts: three payoffs (1 each). 1 pt: preference.

Problem 5 Short Answers [18 points]

(a) True. Let x^*, y^* be optimal and v the value. For every x, $x^\top A y^* \leq v$ because y^* holds Row to at most v; and $x^{*\top} A y^* \geq v$ because x^* guarantees at least v. Hence $x^{*\top} A y^* = v = \max_x x^\top A y^*$.

(b) False. Entry game: an entrant chooses Out $\to (0, 2)$ or In; after In the incumbent chooses Fight $\to (-1, -1)$ or Accommodate $\to (1, 1)$. Then (Out, Fight) is a Nash equilibrium, but Fight is not optimal in the subgame after In, so it is not subgame perfect.

(c) Payoff stream $6, 3, 6, 3, \ldots$:

$$(1 - \delta)\,\frac{6 + 3\delta}{1 - \delta^2} = \frac{6 + 3\delta}{1 + \delta} = \frac{6 + \frac{3}{2}}{\frac{3}{2}} = \boxed{5}.$$

(d) Pure: Player 1's best payoff against L, C, R is $3, 2, 4$; Player 2 picks C, so the pure minmax is $\boxed{2}$. Mixed: Player 2 plays L w.p. $\frac{1}{4}$ and C w.p. $\frac{3}{4}$, giving Player 1 $\frac{3}{4} + \frac{3}{4} = \frac{3}{2}$ from T and $0 + \frac{3}{2} = \frac{3}{2}$ from B. Player 1 can guarantee $\frac{3}{2}$ by playing T and B with probability $\frac{1}{2}$ each (payoffs $\frac{3}{2}, \frac{3}{2}, 2$ against L, C, R), so by the minimax theorem the mixed minmax is exactly $\boxed{\frac{3}{2}}$.

(e) Player 1 has $2^3 = 8$ pure strategies and Player 2 has $3^2 = 9$, so there are $\boxed{72}$ profiles.

(f) False. In Problem 3 with $\delta \geq \frac{2}{5}$, grim trigger is an SPE and C, which is strictly dominated in the stage game, is played in every period on the equilibrium path.

Rubric

(a), (b), (f): 1 pt for the correct True/False, 2 pts for a valid justification.
(c): 1 pt for the discounted sum, 2 pts for the value 5. Giving the unnormalized sum 10: 2/3.
(d): 1 pt for the pure minmax 2; 2 pts for $\frac{3}{2}$ with a mixing strategy for Player 2 (1 pt if the value is asserted without showing that Player 2 cannot do better, i.e. without Player 1's guarantee).
(e): 3 pts for 72; 1 pt for $2 \cdot 3 \cdot 2 \cdot 3 = 36$ or other counts that ignore type-contingent strategies.